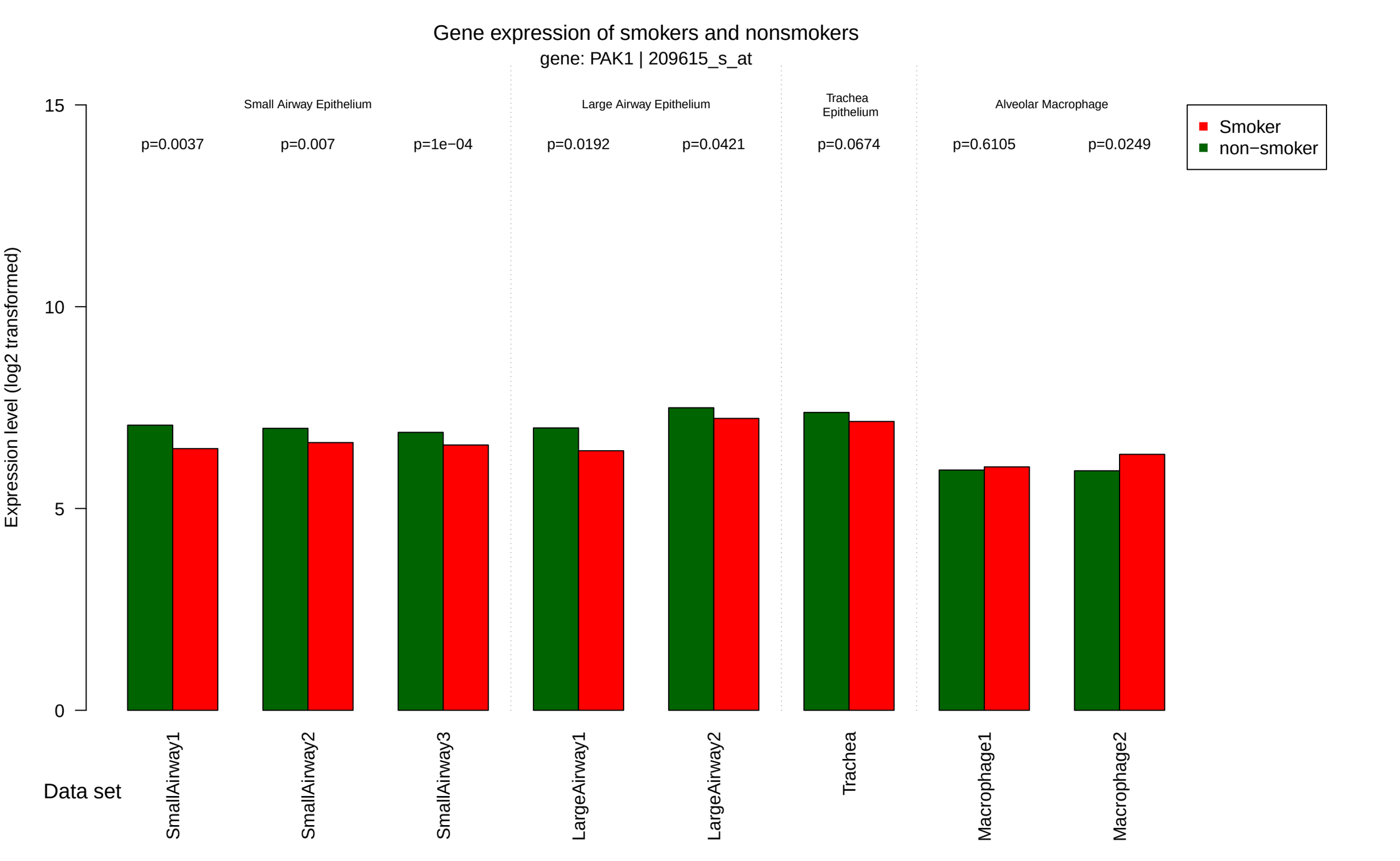
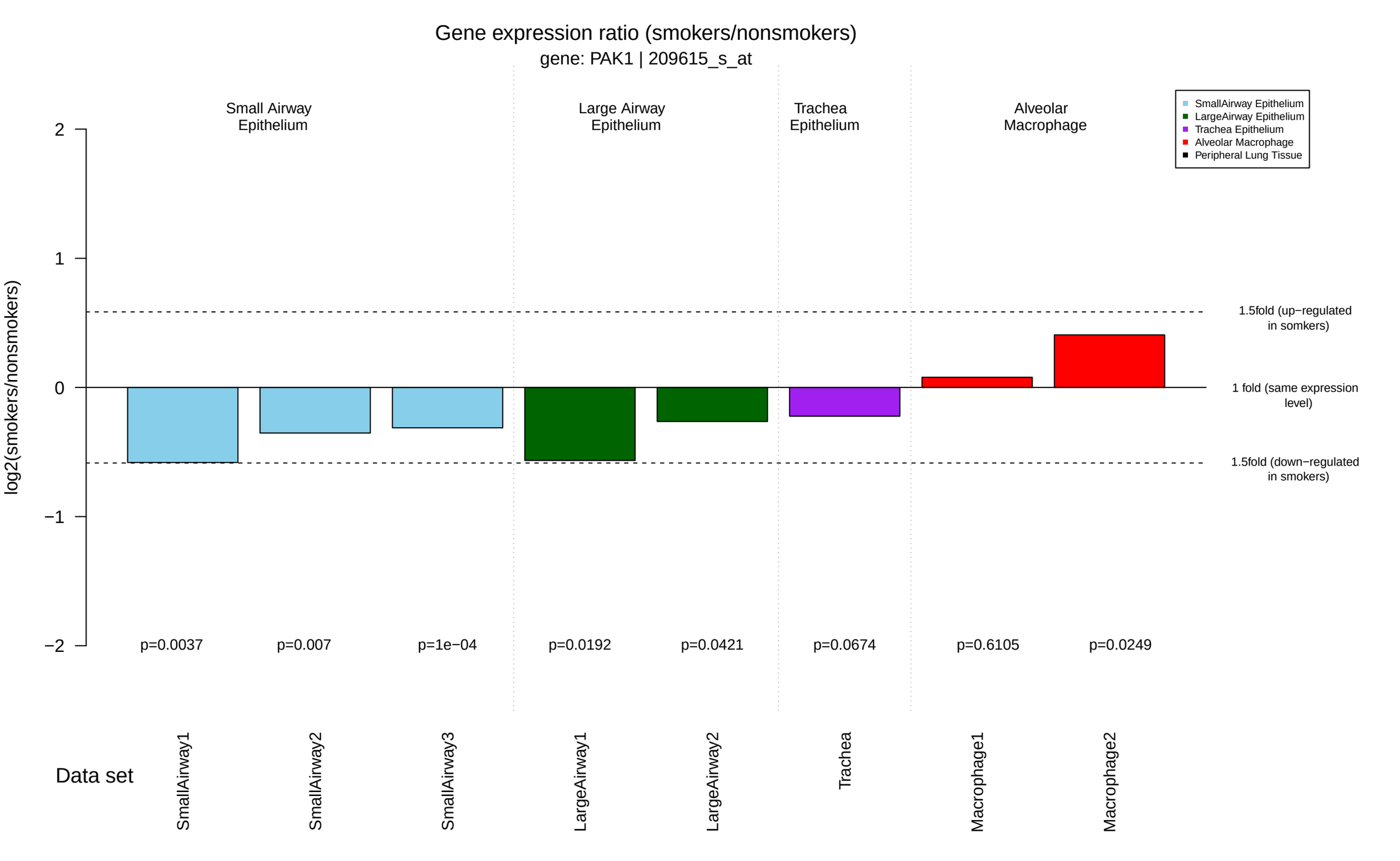
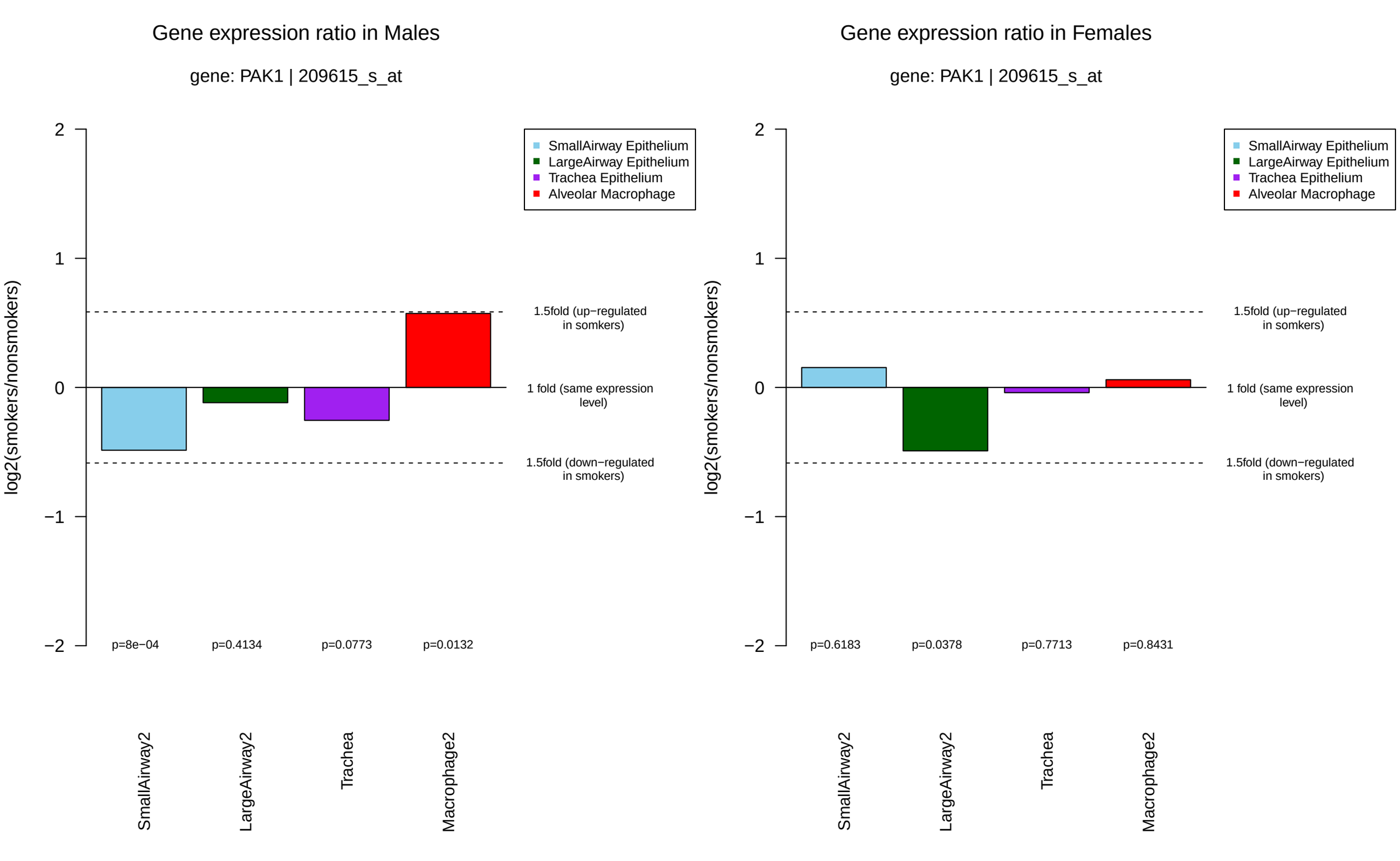
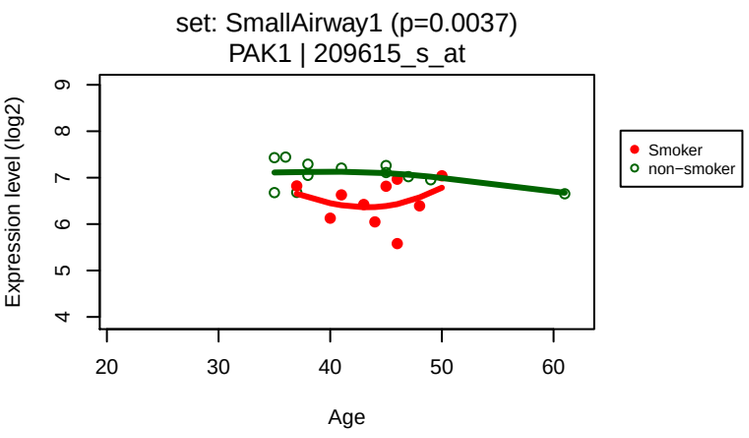
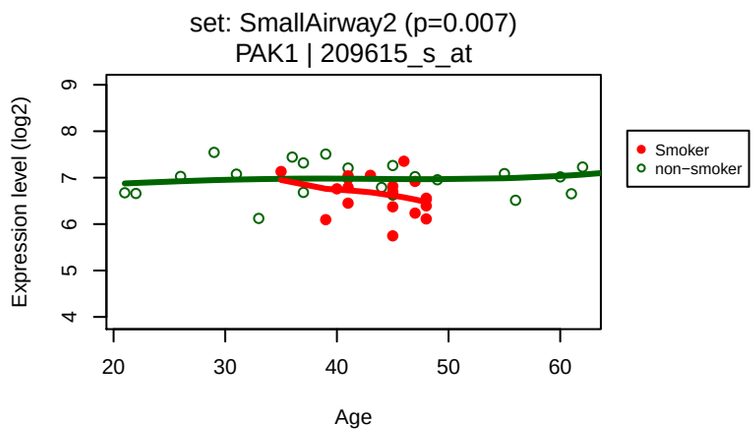
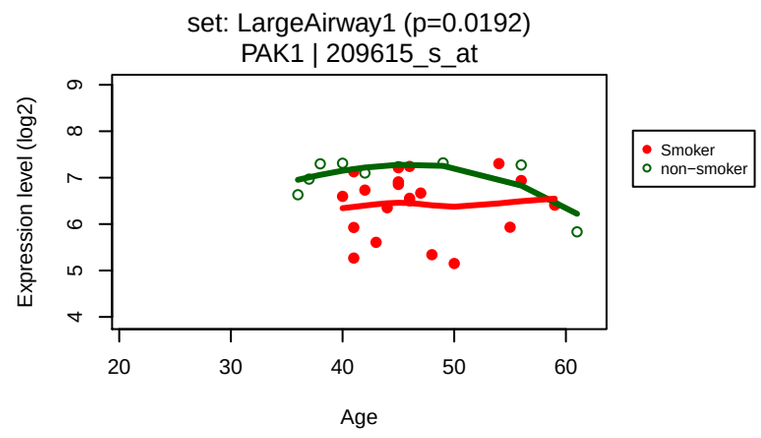

Gene expression of smokers and nonsmokers
gene: PAK1 | 209615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0037
p=0.007
p=1e−04
p=0.0192
p=0.0421
p=0.0674
p=0.6105
p=0.0249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

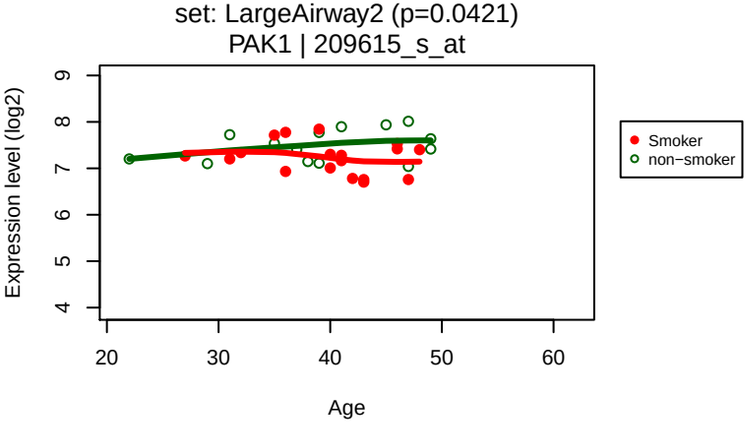

Gene expression ratio (smokers/nonsmokers)
gene: PAK1 | 209615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.007
p=1e−04
p=0.0192
p=0.0421
p=0.0674
p=0.6105
p=0.0249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

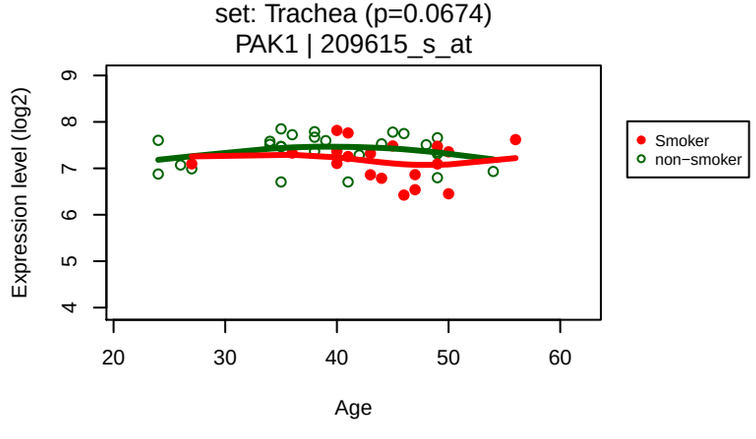

Gene expression ratio in Males
gene: PAK1 | 209615_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=8e−04
p=0.4134
p=0.0773
p=0.0132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK1 | 209615_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6183
p=0.0378
p=0.7713
p=0.8431
SmallAirway2
LargeAirway2
Trachea
Macrophage2

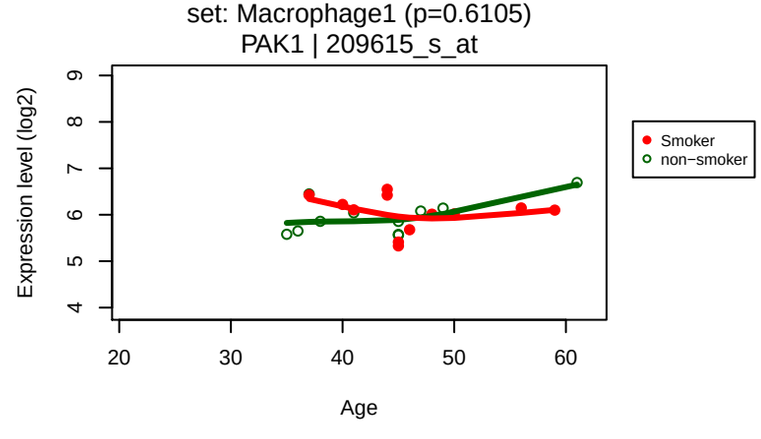

set: SmallAirway1 (p=0.0037)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker

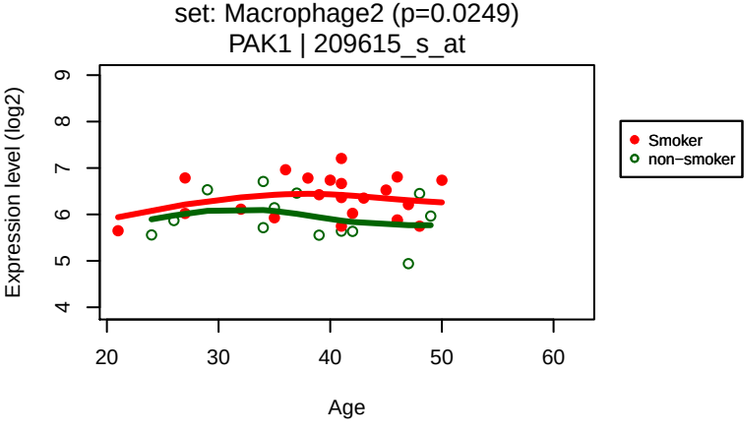

set: SmallAirway2 (p=0.007)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker

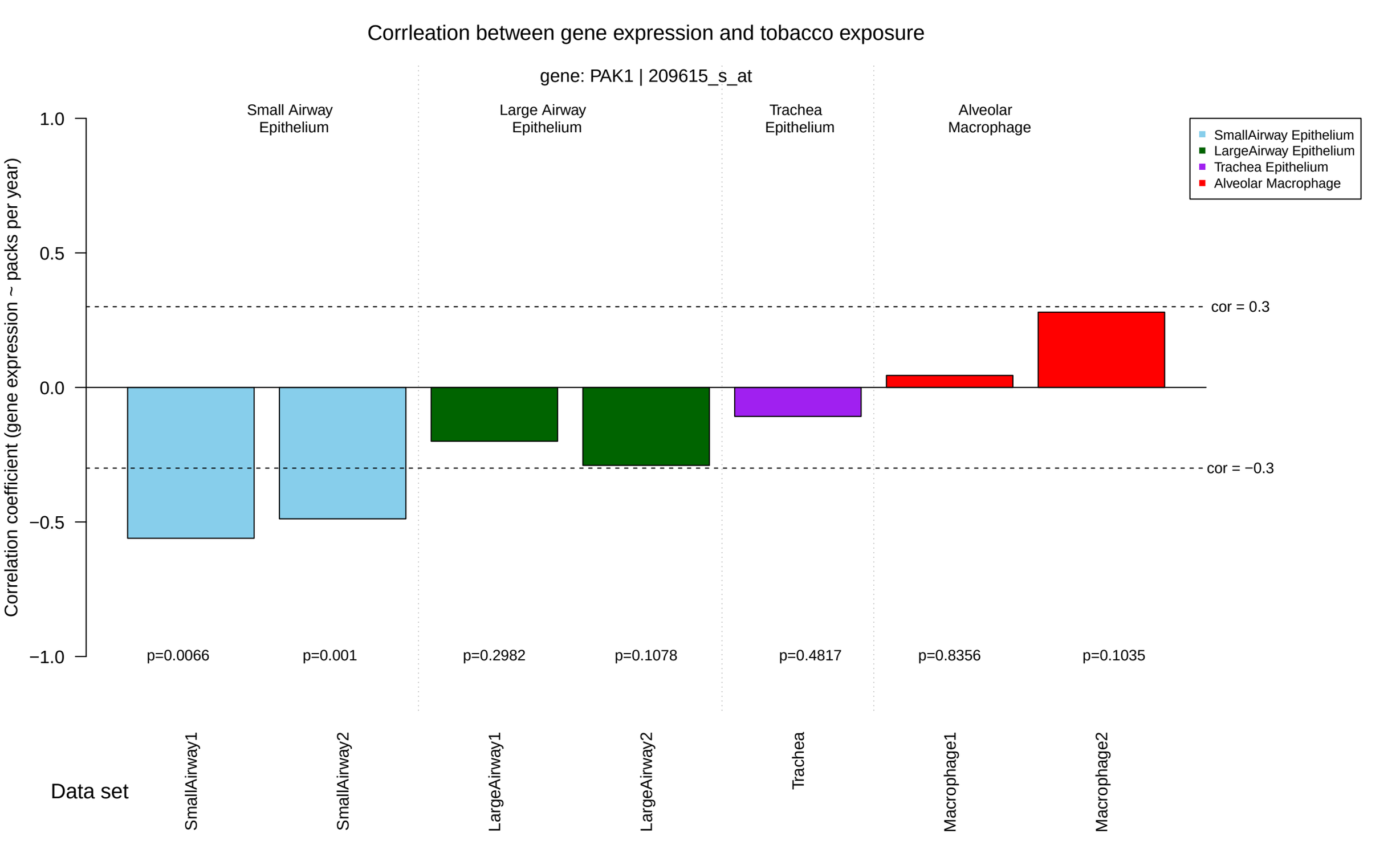

set: LargeAirway1 (p=0.0192)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0421)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0674)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6105)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0249)
PAK1 | 209615_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK1 | 209615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0066
p=0.001
p=0.2982
p=0.1078
p=0.4817
p=0.8356
p=0.1035
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2